

Pricing Bitcoin and Other Assets with Conditional Optimization

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1,2

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Abstract

We propose pricing cryptocurrencies and other assets by measuring the network effects of their adoption. Cong et al. (2020) pioneered the idea of valuation of blockchain as a function of its usage. The idea states that, intuitively, more popular and trusted blockchains are in higher demand and command higher valuations. We extend this model to all financial instruments: cryptocurrencies, US and UK equities, commodity futures and even fixed income. We find that the relationship between prices and their usage is highly nonlinear but also very significant. Using the novel methodology of Aldridge (2024) for non-linear prediction using Conditional Optimization, we generate accurate monthly predictions of cryptocurrency prices that result in 100% excess returns over vanilla buy-and-hold returns ("HDLR"-ing) from January 2022 to December 2024. We test our findings out-of-sample using monthly data. We also apply our methodology to the construction of a diversified minimum-variance portfolio. We show that, according to our model, Bitcoin holdings can be as high as 17% of one's portfolio as long as the changes in volume are actively monitored.

1 Introduction

With Bitcoin (BTC) hitting record prices (USD \$100,000) in December 2024: perhaps the two most popular questions in today's media are:

1. Exactly how to scientifically price cryptocurrencies? and
2. What percentage of a portfolio an institution or an individual should allocate to crypto?

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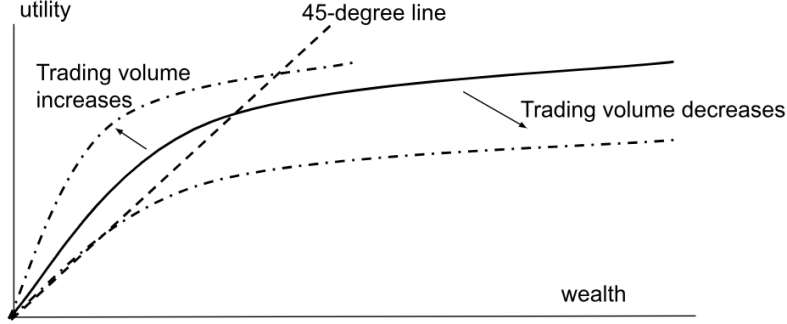


Figure 1: Summary of the proposed model: The slope of the concave utility function shifts with trading volume changes. As trading volume decreases, individuals revise their probabilities of execution downward, becoming more risk-averse. As trading volume increases, individuals become more risk-loving and willing to invest in the asset, driving up prices along the way.

The answers to these pressing questions are just two of several contributions in our paper. To arrive at the results, we extend the conceptual framework of Cong et al. (2020) for the valuation of assets based on their usage. We also draw on the Conditional Optimization of Aldridge (2024) to develop a novel asset pricing methodology for noisy variables that have nonlinear predictors.

Building on the breakthrough ideas of Cong et al. (2020), we propose a simple model of the trading behavior of risk-averse agents that can be applied to all asset classes and crypto in particular. In a nutshell, according to our model, when agents observe trends in trading volume, they calibrate their risk preferences. When the trading volume increases, the agents become more relaxed about their future chances of trade execution and, therefore, less risk averse. The updated risk preferences change the shape of the agents' utility functions, making them less concave when the trading volume increases and more concave when the trading volume decreases (Figure 1). The updated risk preferences filter into the agents present valuation of future payoffs and into the price the agents are willing to pay for the asset. As agents feel more confident about being able to sell the asset in the future, their probabilities of positive payoffs increase, and the price they are willing to pay for the asset also increases.

We estimate that, in the traditional minimum-variance framework, the optimal portfolio holding of Bitcoin (BTC-USD) using the latest data as 3%, up from 2% estimated by Liew and Hewlett (2017). However, we find that holding stablecoins such as Tether (USDT) is even more beneficial for one's portfolio and can optimally take as much as 90% of one's portfolio. We hypothesize that the probable reasons for a large Tether allocation are 1) high traded volume, 2) exposure to crypto, yet 3) relative stability and low volatility as compared

to, say, Bitcoin and other cryptocurrencies. However, using the novel Conditional Optimization approach of Aldridge (2024), we show that Bitcoin can be as much as 17% of one’s portfolio, provided that the changes in Bitcoin volume are tracked on the monthly basis.

Cong et al. (2020) were the first to hypothesize that the price of digital platforms is related to their use. According to Cong et al. (2020), the higher the “productivity” of a given digital platform or token, the higher the valuation of the platform. In this paper, we interpret the productivity of crypto currencies as their transactional power manifested by their trading volume. Cong et al. (2020) further showed that the price of a digital platform is nonlinear with respect to its adoption and follows an S-shaped curve. In this paper, we extend the argument of Cong et al. (2020) to show that the prices of Bitcoin and other digital currencies are a function of currency usage, i.e., volume traded. We also test other assets, like US, European and Asian equities, traditional currencies, and commodity futures, and find that trading volume also predicts the returns on those assets as well.

In our empirical study with monthly data, we find that the returns of cryptocurrencies have a quadratic dependency on the realized changes in the traded volume. Returns on other assets, such as traditional equities, are linearly dependent on changes in volume. The quadratic fit supports our hypothesis of concave utility-based pricing of cryptocurrencies.

Using the trading volume as predictors and their functional specification, we derive a new technique for cryptocurrency pricing. The technique is based on Conditional Optimization developed by Aldridge (2024). The process uses a loss concept from the machine learning literature to optimize pricing for nonlinear predictors.

We further empirically demonstrate the predictive power of the Conditional Optimization methodology in crypto pricing. Based on history, the methodology allows us to probabilistically fix future realized values of a given noisy variable that is a polynomial function of some predictor available today. Based on the Conditional Optimization technique, we can make much more accurate predictions of the realized prices in the future.

We also show a step-by-step price calculation process using the technique. Empirically, we find that past trading volume is a significant factor in asset returns across all assets. These results conform to the joint volume and return modeling by Lo and Wang (2006).

In this paper, we aim to make the following contributions to the discourse.

1. We document that, consistent with Cong et al. (2020), the trading volumes of cryptocurrencies are significant predictors of the prices of cryptocurrencies.
2. We extend Cong et al. (2020)’s idea to other financial markets to show that the returns on most financial instruments can be predicted using trading volume, consistent with Brennan and Subrahmanyam (1996), Brennan et al. (1998), Lee and Swaminathan (2000), Lo and Wang (2006) and Aldridge and Robinson (n.d.).

3. We derive a new methodology for pricing assets with nonlinear factors using Conditional Optimization (CO) of Aldridge (2024).
4. Using the Conditional Optimization framework by Aldridge (2024), we propose a portfolio optimization approach that incorporates nonlinear predictors to create the most stable portfolio outcomes.
5. We empirically compare the prediction performance of the proposed CO methodology and document the positive performance for the prediction a month ahead. This approach is beneficial to large institutional investors, such as pension funds, mutual funds, and hedge funds, as well as small investors seeking to make informed decisions about asset allocation.

2 Related Literature

2.1 Asset Pricing

The core of our methodology is based on Cong et al. (2020) and diverges considerably from the traditional risk-return literature like Breeden (1979) and others summarized in Cochrane (2005) and Fabozzi (2008). However, we lean on 1) the fundamental pricing literature like van Binsbergen and Opp (2019), Cho and Polk (2024) and others who perfect pricing based on future cash flows, and 2) risk aversion of agents as in Jr. and Chow (1992) and many others. Separately, recent work on asset price efficiency like Bai et al. (2016) helps explain why the performance results of our model are so significant.

2.2 Factorizing Returns

Return factorization goes back to the seminal Capital Asset Pricing Model (CAPM) of Sharpe (1964) and Lintner (1965). As discussed in Harvey et al. (2015), Feng et al. (2020), and others, the factor pricing literature exploded over time. The volume of trades appears as a factor in Lo and Wang (2006) and Aldridge and Robinson (n.d.).

Most of the factorization literature is linear.

2.3 Nonlinear factorization

Most of the factors discussed in the now traditional "factor zoo" of the financial literature (Harvey et al. (2015)) are linear in nature. a Nonlinear factor models have been implemented in the literature by one or more of the following means.

- Kernel smoothing
- Neural networks
- Other Machine-Learning Strategies

The neural network approach was pioneered by Levin (1995). Gu et al. (2020) provide a summary and comparison of various machine learning methods in the context of risk premia prediction.

Gao et al. (2022) deploy Support Vector Machines to find the right mix of stocks.

2.4 Volume in Asset Pricing

Brennan et al. (1998) were the first to document volume as a factor in asset pricing. Lo and Wang (2006) used trading volume to find optimal allocations for hedging portfolios. However, researchers have long used the traded volume for related purposes. For example, Brennan and Subrahmanyam (1996) and Almgren and Chriss (2000) used volume as a proxy for liquidity.

Lee and Swaminathan (2000) show that the volume of trading is correlated with price momentum in equities. They found that while both "winner" and "loser" portfolios experience high trading volume, the trading volume for winners is higher than that for losers. Most recently, Aldridge and Robinson (n.d.) identified trading volume as the leading factor in pricing futures.

2.5 Portfolio Optimization

It is really difficult to do justice to the numerous advanced literature on Portfolio Optimization. Markowitz (1952) pioneered optimization in portfolio allocation. Clarke (2006) introduced minimum-variance portfolios that can be viewed as an efficient computation of the optimal mean-variance framework.

The most recent work includes DeMiguel et al. (2024) who use volatility as the conditioning variable in their portfolio construction. Pani and Fabozzi (2022) studied changes in corporate variables as a cause of portfolio momentum. Barroso and Detzel (2021) study the impact of transaction costs, arbitrage risk, and short-sale impediments on portfolio returns. Moreira and Muir (2017) include value, momentum, profitability, return on equity in their portfolio optimization. Earlier studies incorporating factors into portfolio management included Rishel (1999), Jacobs et al. (2005), and many others.

Lo and Wang (2006) were the first to use the trading volume in portfolio construction. However, Lo and Wang (2006) opted for a linear relationship between trading volume and portfolio allocation.

2.6 Cryptocurrency Pricing

In addition to Cong et al. (2020), several researchers considered the economics of Bitcoin. In the early days of Bitcoin, Yermack (2015) was pessimistic about the prospects of the cryptocurrency. Recently, Huberman et al. (2021), Liu and Tsyvinski (2020), and Sockin and Xiong (2023) have made an argument for Bitcoin as a transaction medium valuable for its network effects.

Bouri et al. (2017) and Huang (2024) found that cryptocurrencies are a useful diversifier in traditional portfolios. Bouri et al. (2017) found Bitcoin is a poor

investable asset. However, Glaser et al. (2014) found that most users of Bitcoin were in it for investment opportunities.

The relationship of Bitcoin prices and Bitcoin trading volume was considered by Balcilar et al. (2017), who found that to be indeed non-linear.

This paper is also related to the literature on asset pricing and, in particular, to the work on cryptocurrency trading. At the time this paper was written, much of the latest research on cryptocurrency pricing developed along the following lines:

- momentum-based strategies
- factors
- triangular arbitrage
- investor attention

The literature on momentum-based pricing of cryptocurrencies is perhaps the most voluminous. Christian Fieberg and Zaremba (2023) document that of 3,900 coins, past winners consistently outperform losers. Liu et al. (2022) and Jia et al. (2022) find strong momentum in the smaller currencies. Proelss et al. (2025) find that momentum strategies work better when the volatility of the market is high, as measured by VIX. Corbet et al. (2019) document that technical strategies, such as momentum, among others, work in crypto markets. However, Grobys and Sapkota (2019) showed that from 2014 to 2018, momentum was not significant in their sample of 143 cryptocurrencies. Kogan et al. (2024) found that momentum strategies in crypto are primarily initiated and perpetuated by retail investors.

A strand of literature focused on exogenous factors driving cryptocurrency returns. Kristoufek (2015) found that the Chinese market influences Bitcoin prices. Hu et al. (2019) showed that most of the 220 currencies they considered were influenced by the price of Bitcoin. Assamoi et al. (2025) showed that cross-sectionally, cryptocurrencies have strong relationships with factors related to equities, foreign exchange, precious metals and market volatility. Ahmadova et al. (2024) and Akbari et al. (2024) further found that cryptocurrency returns are positively correlated with equities and commodities and negatively correlated with foreign exchange. Papadamou et al. (2021) found that gold became a more influential factor during bear markets.

Huang et al. (2022) develop a triangular arbitrage model between BTC and fiat currencies. Huang et al. (2022) show that triangular arbitrage opportunities are particularly pronounced when market volatility is high.

Liu and Tsyvinski (2020) found that the returns of cryptocurrency are positively related to investor attention. Lucey et al. (2022) develop a news index that helps capture the volatility of cryptocurrencies.

The theoretical Conditional Optimization methodology of Aldridge (2024) is related to the Predict-then-Optimize literature of Ban and Rudin (2013), Bertsimas and Kallus (2020) and Elmachtoub and Grigas (2022), among others.

As discussed in section 3 of this paper, Aldridge (2024) develops a theoretical framework for closed-form optimal decisions based on an uncertain variable. When the variable is a polynomial function of a known predictor and the degree of the polynomial is known, Aldridge (2024) showed that the best decision is made based on the expected value of the variable calculated with a polynomial one degree lower than the original functional form. When the degree of the polynomial is not known, Aldridge (2024) and Brokmann et al. (2024) show that the optimal decision can be approximated by a linear prediction or a machine learning estimate. Aldridge (2024) shows that when the polynomial degree of the variable versus the predictor is not known, linear prediction works best when the actual polynomial degree is low, and machine learning estimation works best in highly non-linear cases.

Our work is also related to the econometric literature for fitting nonlinear functions. The papers in this field include Hamilton (2001), Schennach (2004), Andrews and Mikusheva (2014) and Sadeghi Eshkevari et al. (2022) among others. Hamilton (2001) uses diffusion to find the shape of the nonlinearities. Schennach (2004) rely on the Fourier transform to estimate the nonlinear functional specifications. Andrews and Mikusheva (2014) adapt Maximum Likelihood to fit the nonlinearities of data. Sadeghi Eshkevari et al. (2022) propose a neural network approach.

A separate stream of literature focuses on cryptocurrency market design and the pricing that can be extracted from the market structure. For example, Capponi et al. (2023) investigate how the volume in private pools on the blockchain impacts currency pricing.

3 Methodology

3.1 A Simple Utility Model

We consider a simple utility model with a concave utility function of wealth. A concave utility function models the preferences of an individual i who is risk averse and prefers stable outcomes to risky ones. Our simple argument is that if the levels of trading volumes are persistent over time, then higher trading volumes ensure that we can transact in the future. In other words, high trading volume eases market frictions and ensures that if we get into a position today, we will be able to successfully liquidate our investment at a future date.

3.1.1 Extension of Cong et al. (2020)

Mathematically, users of an asset are seeking affirmation that, in the future, their today's investment will have eager buyers. Cong et al. (2020) express this idea as the following utility function:

$$du_{it} = x_{it}^{\alpha} (V_t A_t e^{\theta})^{deg} dt - \phi t - x_{it} r dt \quad (1)$$

where:

- x_{it} represents agent i 's holdings of the given asset at time t . In our model, we assume that x_{it} is independent of the actions of other users, V_t and ΔV_t . For example, x_{it} may represent the wealth of agent i at time t .
- V_t is the asset's user base at time t
- A_t is the platform's productivity at time t . According to Cong et al. (2020), a positive shock to productivity increases the user base and a negative shock does the opposite.
- θ is the user type. We depart from Cong et al. (2020) by assuming a homogeneous pool of users. θ is constant for all our agents.
- $\alpha \in (0, 1)$ is a constant specific to each asset, which Cong et al. (2020) call the "productivity constant" because it reflects the functionality, regulatory environment, and other factors of the given asset.
- deg is an exponent of the trading volume. V^{deg} multiplier on the power utility reflects the preferences of agents for larger, more liquid markets, $deg = 1 - \alpha$ in Cong et al. (2020) for computational convenience. However, in our model, we relax this assumption and let deg be any positive number.
- ϕ is the agent's unit cost participation on any platform. This can reflect the time involved in transacting.
- Finally, r is the opportunity cost per unit of time of using a specific platform.

In our model, we strictly focus on how user preferences evolve with changes in other users' participation, that is, the trading volume. Therefore, we propose a much simplified representation of equation (1):

$$du_{it} = \gamma x_{it}^\alpha (V_t)^{deg} dt \quad (2)$$

Equation (2) can be interpreted as the marginal utility of agent i (with wealth x_{it}) derived from holding an asset with trading volume V_t at time t . Marginal utility can be interpreted as the price an agent is willing to pay for an additional unit of the asset under consideration. We will therefore use equation (2) in calculating the prices of assets.

Theorem 1. *As the trading volume in a particular asset decreases, agents become more risk-averse. In contrast, as the volume of trade on the asset increases, the agents become more risk-loving.*

Proof. A decrease in the trading volume, $\Delta V_t < 0$, in the utility specification of equation 2 decreases the utility of agent u_{it} at all wealth levels x_{it} , resulting in more risk-averse preferences and appetite for investment in the asset. In contrast, when $\Delta V_t > 0$, the utility u_{it} increases for all wealth levels x_{it} , causing agents to become more risk-loving and take greater chances of investing in the given asset. $\square$

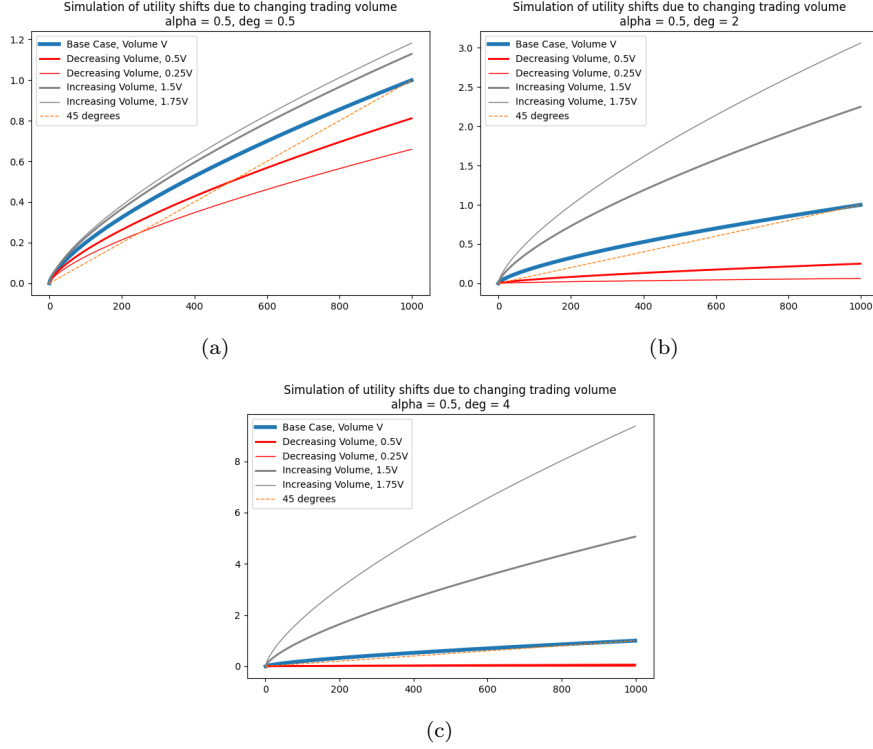


Figure 2: Simulation of agent utility under various polynomial specifications over trading volume.

3.1.2 Simulation

In this section, we verify the hypothesis of our model via a basic simulation. To do so, we return to equation (2). Since none of the variables in equation (2) are functions of t , we can rewrite it as a utility function with the following specification:

$$u_{it} = \gamma x_{it}^\alpha V_t^{deg} \quad (3)$$

where V_t is the volume of the asset traded at time t . Next, we test our model with increases and decreases in V_t and various values of deg .

Next, we consider how the utility function changes as a function of changes in the volume traded. Figure 2 shows the base-case utility function with α and some nominal value of trading volume, V . The figure also plots changes in the utility function with increases and decreases in trading volume ΔV . As V decreases, the agent becomes more risk-averse.

In Figure 2, the 45 degree line shows the risk-neutral zone of the agent. Negative changes in trading volume make the utility function more concave, that is, agents become more risk-averse. Positive changes in trading volume tilt

agents' utility to risk-loving. The higher the exponent $\deg$ in $V^{\deg}$, the more drastic the change in the utility function.

As the utility function shifts, so does the agent's appetite for uncertain future payoffs. When $\deg$ is low, the agent follows his standard routine with slight preference for larger trading volumes. When $\deg$ increases, the agent changes preferences considerably and jumps to assets with high $\deg$, but only when the change in trading volume is positive. When the trading volume decreases, so does the agent's utility across all wealth levels, and the agent chooses to preserve his current wealth over speculation.

In other words, agents are more likely to take a chance on a given asset when asset circulation (trading volume) increases. In contrast, agents are more likely to dispose of the asset when the asset user base (again, trading volume) is shrinking.

3.1.3 Network Effects in Asset Pricing – A Bayesian Perspective

The findings make sense in the context of traditional economics. The price of an investment can be viewed as the present value of the future payoff. If today we buy the asset for some price $P_{it} > 0$, we want to make sure that at some future time $t + 1$, we can sell the asset for some price $P_{i,t+1} > 0$. Today, we expect this asset to be worth $E_t[P_{i,t+1}] > 0$. The expected value of our asset is directly dependent on the existence of the network of eager buyers of our asset, that is, the trading volume.

Theorem 2. *Future trading prices are dependent on the network effect: agents' collective desire to trade a given asset in a specific market.*

Proof. A risk-averse agent i will buy an asset today only if he is certain that he can sell it tomorrow, even if at a loss. To sell the asset tomorrow, the market for the asset needs to have a buyer j who believes that she will be able to sell the asset in the following days, and so on. Let's assume that each agent always trades just one unit of the same asset in one market. Then we can express the probability that the agent enters the market as follows:

$$P_{it}(\text{buy}_i) = P_{i,t+1}(\text{sell}_i) = P_{j,t+1}(\text{buy}_j) = P_{j,t+2}(\text{sell}_j) = \dots \quad (4)$$

If the network size of agents trading a given asset stays constant, so will the price of the asset. Now, let us introduce information. Suppose that at time t each agent can observe the previous trading volume V_{t-1} . Past trading volumes are generally observable today in all markets except foreign exchange.

When agent i observes the trading volume V , he updates his belief about his future execution probability. If the trading volume increases over time $\Delta V > 0$, the agent assumes that the volume will continue to grow.

$$\Delta V_{t+1} = \sum_j \mathbf{I}_{j,t+1}(\text{buy}_j) - \sum_i \mathbf{I}_{i,t}(\text{buy}_i) \quad (5)$$

Notice that we are counting buy trades only and for two reasons:

1. Every buy trade needs a corresponding sell trade to complete. Thus, counting buy trades implicitly counts sell trades as well.
2. Buy volume is more important for liquidation of assets, especially in the times of crisis when everyone sells.

In a Bayesian formulation, we can represent such a dynamic as follows.

$$P_{k,t+2}(buy_k|\Delta V_{t+1} \geq 0) = P(\Delta V_{t+1} \geq 0|buy_k)P(buy_k)/P(\Delta V_{t+1} \geq 0) \quad (6)$$

The agents' probabilities change with all the agents' collective actions. If the trading volume increases, $P(\Delta V_{t+1}|\text{buy}_k) \geq 0$, so is the network trading the asset and the resulting desire of other agents to participate in the transactions. In other words, the future trading volume is dependent on the agents' collective desire to trade a given asset in a specific market. At the same time, the probability that future trades will take place is a function of positive changes in the volume of trade. Thus, each agent's valuation of the asset is automatically conditioned on the trading volume of the asset. $\square$

3.2 Optimal Pricing and Portfolio Construction

3.2.1 Optimal Price Prediction

Here, we use Conditional Optimization of Aldridge (2024) to predict the prices of an asset with polynomial predictors.

Our decision at time t is the optimal portfolio allocation, a vector of w_t^* , corresponding to a vector of future returns R_{t+1} . The optimality of our decision is measured at time $t + 1$. At time t , the $t + 1$ returns on the assets, R_{t+1} , are not observable; however, the predictors X_t can be observed. Following now-standard machine learning approach, we can measure the *suboptimality* of our decision w_t^* as a loss function between the $t + 1$ portfolio value expected at t and the realized portfolio value:

$$Loss = E_t[w_t' R_{t+1}|X_t] - w_t' R_{t+1} \quad (7)$$

Per Aldridge (2024), at time t our objective is to find the optimal w_t^* to minimize the Loss. Since we fix w_t at time t ,

$$E_t[w_t] = w_t \quad (8)$$

and the Loss function can be rewritten as:

$$Loss = E_t[w_t' R_{t+1}|X_t] - E_t[w_t]' R_{t+1} \quad (9)$$

Next, the expected value of Loss is:

$$E_t[Loss] = E_t[w_t' R_{t+1}|X_t] - E_t[E_t[w_t]' R_{t+1}] \quad (10)$$

$$E_t[Loss] = E_t[w_t' R_{t+1}|X_t] - E_t[w_t]' E_t[R_{t+1}] \quad (11)$$

We can recognize the latter expression for Loss as the covariance between time- t portfolio allocation and asset returns at time $t + 1$. Since the portfolio weights are the same at the decision time t and the return observation time $t + 1$, Loss can be more generally rewritten as:

$$E[Loss] = cov(w^*, R) \quad (12)$$

Conditioning on predictors observable at time t , we find can rewrite the Loss as:

$$E[Loss|X] = cov(w^*, R|X) \quad (13)$$

where w and R are vectors of portfolio weights and returns, respectively, and X is the matrix of predictors. Generally, predictors can be any data related to future asset returns. According to the discussion in this paper, predictors can be trading volumes and degrees of nonlinearity.

Following Aldridge (2024), to minimize Loss, we need to minimize covariance from equation (12). To do so, Aldridge (2024) proposes to use first-order conditions computed with respect to X . Aldridge (2024) shows that when future returns are a polynomial function of predictors, $R_{t+1} \sim X_t^{deg}$, and the decision variable w_t is linear in R , then the optimal decision w_t is based on $R \sim X_t^{deg-1}$, a derivative of R with respect to X :

$$w_t^*(R_{t+1}) = w_t^*(X_t^{deg-1}) \quad (14)$$

In this paper, our predictor is the change in volume traded, ΔV_t . Suppose returns are related to ΔV_t as

$$R_{t+1} \sim (\Delta V)^{deg} \quad (15)$$

Then our optimal portfolio allocation vector w_t^* should be computed using

$$R = (\Delta V)^{deg-1} \quad (16)$$

3.2.2 Minimum-Variance Portfolios: Conditional Optimization

The traditional minimum-variance allocation solution (Clarke (2006)) stipulates that:

$$w = \frac{\Sigma^{-1}e}{e\Sigma^{-1}e} \quad (17)$$

where $\Sigma = cov(R)$ is the covariance matrix of asset returns.

As before, $R \sim \Delta V^{deg}$. Using a standard prediction specification,

$$\Sigma = E[\Delta V' \Delta V]^{deg} - E[\Delta V]^{2deg} = Var[\Delta V^{deg}] \quad (18)$$

However, according to Aldridge (2024), the optimal polynomial degree for prediction is $deg - 1$. Therefore, in the Conditional Optimization (CO) specification, we have the following variance-covariance matrix:

$$\Sigma_{CO} = E[\Delta V' \Delta V]^{deg-1} - E[\Delta V]^{2(deg-1)} = Var[\Delta V^{deg-1}] \quad (19)$$

The optimal weights for our Conditional Optimization portfolio, therefore, are:

$$w_{CO} = \frac{\Sigma_{CO}^{-1}e}{e\Sigma_{CO}^{-1}e} \quad (20)$$

where Σ_{CO} is the covariance of changes in returns of assets raised to the power of $deg - 1$.

4 Estimation

We conducted a series of empirical studies to better understand the relationship between trading volume and subsequent cryptocurrency valuation measured by crypto returns.

4.1 Fitting a Polynomial

First, we fit the crypto returns R_i as a polynomial function of volume V : $R_i \sim V^{deg}$. To do so, we use polynomial regression. For various values of deg on in-sample data, we estimate the optimal coefficients $\{p_i\}$ for a polynomial representation of V^{deg} that best fit R_i :

$$R_i = p_{deg}V_i^{deg} + p_{deg-1}V_i^{deg-1} + p_2V_i^2 + p_1V_i + p_0 \quad (21)$$

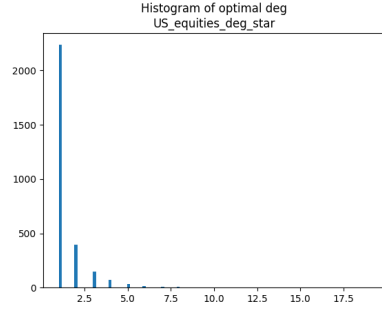
Next, to find the best-fit polynomial degree, deg^* , we reconstruct the returns $\hat{R}_i$. To do so, for every degree $deg \in [1, ..., 40]$, we compute $\hat{R}_i$ using the polynomial function of equation (21). We next measure the fit of each deg as the Adjusted R^2 of regressions of R_i on $\hat{R}_i$. We conducted two sets of regressions: 1) on the "in-sample" data (before January 1, 2022), and 2) on the "out-of-sample" data (between January 1, 2022, and December 31, 2024). We found that in-sample, the precision of the model continues to increase with the increase in deg , as expected from Taylor's decomposition. However, out-of-sample, the highest Adjusted R^2 was obtained for small deg , and the Adjusted R^2 subsequently declined for higher deg values. Thus, for each financial instrument considered, we picked and reported deg with the maximum Adjusted R^2 out-of-sample. To verify our findings, we examine deg in different time periods in Section 4.5.

Figure 3 shows the distribution of the optimal deg^* for different financial instruments. Figure 4 shows the distribution of Adj R^2 used to measure the goodness of fit of deg^2 .

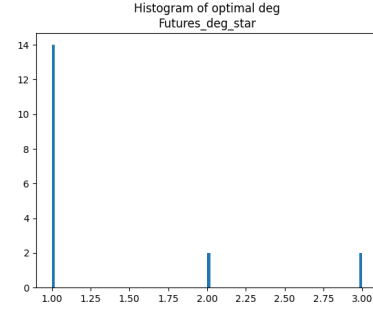
4.2 Estimating Dependency

Next, we find the empirically optimal deg . To do so, we run out-of-sample linear regressions of R_i on $\hat{R}_i$ estimated from equation (21) and identify the closest fit. This step provided us with evidence of the polynomial relationship, degree deg , between volume and returns.

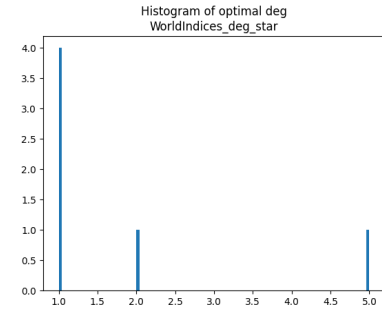
Common wisdom and, according to DeMiguel et al. (2007), even the Babylonians in the 4th century AD suggested splitting an investment portfolio into



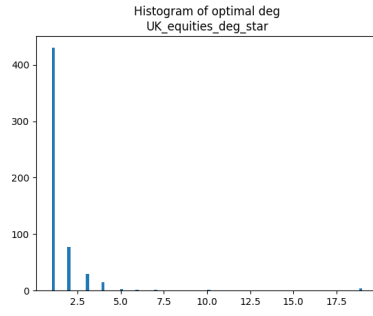
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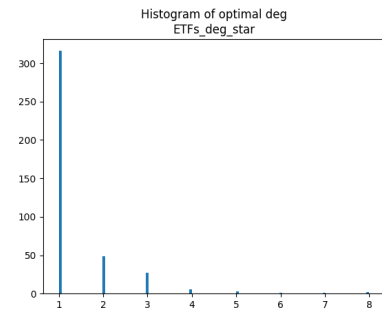
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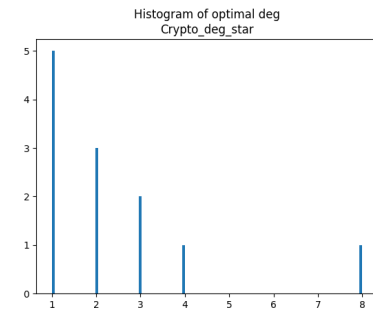
(c)



(d)

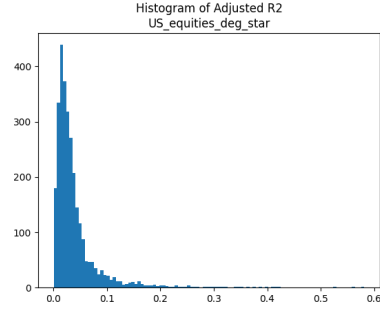


(e)

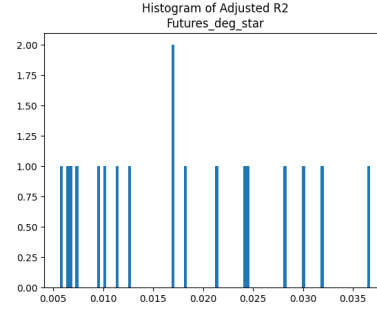


(f)

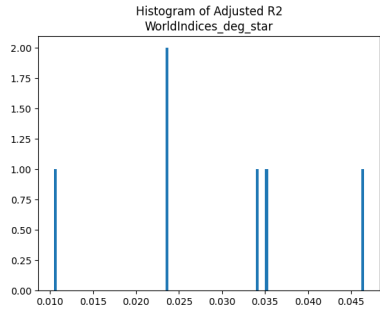
Figure 3: Distribution of Optimal polynomial deg^* for the out-of-sample approximation $R_i = degV^{deg^*}$ for Different Asset Classes. The goodness of fit is measured by the Adj. R^2 .



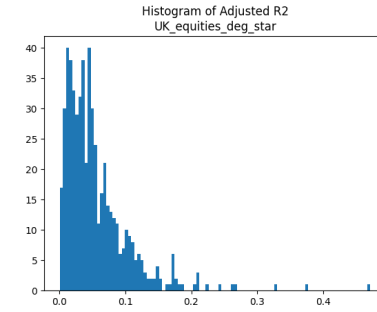
(a)



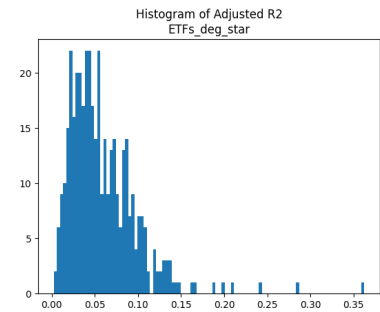
(b)



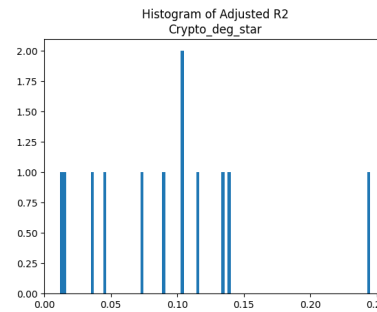
(c)



(d)



(e)



(f)

Figure 4: Distribution of Adj. R^2 for the out-of-sample linear regression $R_i = degV^{deg*}$ for Different Asset Classes

at least three parts. In our experiment, we consider four different assets: U.S. equities represented by the S&P 500 ETF (SPY), U.S. Treasuries proxied by bond futures (ZB=F) and two types of cryptocurrencies: Bitcoin and USD Stablecoin Tether. Using a rolling window estimator, we calculate the coefficients of a polynomial regression, equation (21), for different values of $deg \in [0, 40]$ for Bitcoin (BTC-USD), S&P500 ETF (SPY), USD stablecoin Tether (USDT-USD) and US Treasury Futures (ZB = F). Treasury futures were selected to represent US bonds in the optimal portfolio.

Once we found the coefficients in the sample for the polynomial fit of equation (21), we roll our estimation window forward one month and use the coefficients for the following months. We then find the deg with the best fit in the next month (out-of-sample). The degree deg with the best fit out-of-sample becomes our optimal degree for the following month.

Figure 5 shows the optimal polynomial degrees for each instrument over time. Although USD Tether (USDT) and the S&P500 ETF (SPY) remained largely linear with respect to changes in their volume, Bitcoin (BTC) has been mostly quadratic in ΔV until December 2022 and then became cubic from 2023. The U.S. Treasury futures displayed a most remarkable shifts in their polynomial degree of ΔV from 1 to 5.

4.3 Evaluating Prediction

Next, we let the tires hit the road and apply our techniques to the actual portfolio construction. We then test our portfolio performance and compare it with the base case of portfolio management based on asset returns only, without factors.

To reflect the preferences of institutional investors, we conducted our analysis using monthly price and volume data. To do so, using the rolling window set up introduced in Section 4.2, we construct a series on minimum-variance portfolios following the discussion in Section 3.2.2. The in-sample vs. out-of-sample timeline within each window is shown in Figure 6. Each "estimation window" shown in Figure 6 is then shifted one month forward to simulate yet unknown market conditions.

Table 1 shows the results of the out-of-sample estimation of equation (21) for different polynomial degrees deg . As Table 1 shows, the average optimal deg^* fell between 1 and 2 (linear and quadratic) for most financial instruments, except crypto. For cryptocurrencies, the average deg^* was 2.42, falling between quadratic and cubic (the latter S-shaped as predicted by Cong et al. (2020)).

Separately, we also calculated the monthly out-of-sample returns on the strategy following the Conditional Optimization technique as follows. For each financial instrument i , once we estimated the optimal degree deg_i , we made a Conditional Optimization return forecast for the following month:

$$\hat{R}_{CO,i,t+1} = \Delta V_{it}^{deg_i-1} \quad (22)$$

We then created a simple algorithmic trading strategy:

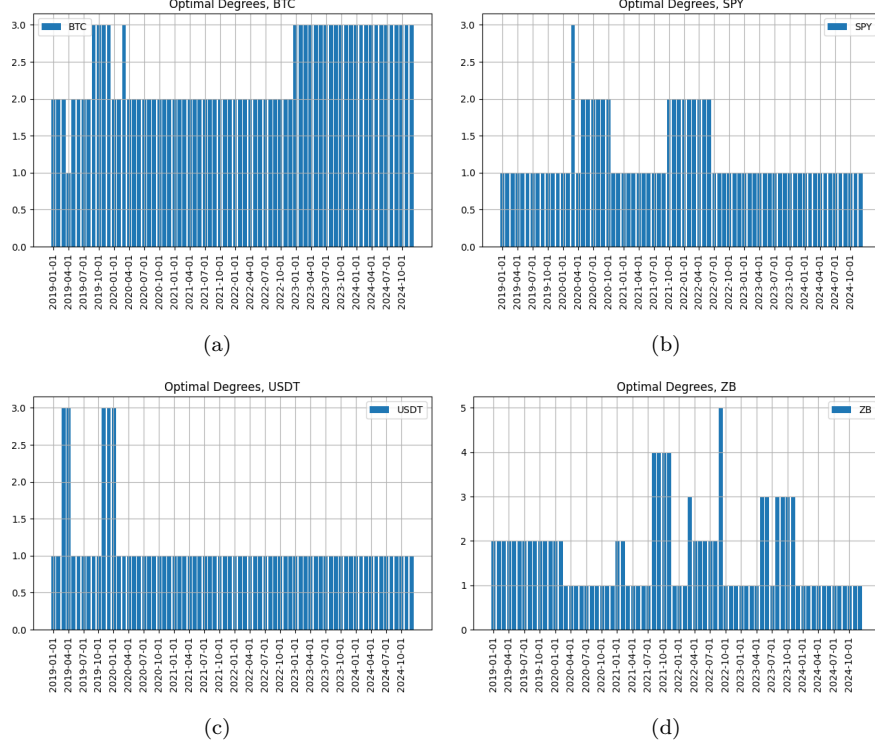


Figure 5: Optimal polynomial degrees of $R \sim \Delta V^{deg}$ for Bitcoin (BTC-USD), the S&P 500 ETF (SPY), USD-Tether (USDT-USD), Treasury Bond Futures (ZB=F), with 36-month in-sample estimation period preceeding each given date.

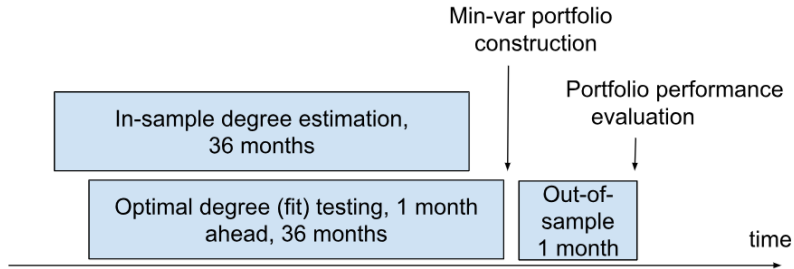


Figure 6: A snapshot of a "Window" in the rolling window performance evaluation process of Conditional Optimization of minimum-variance portfolios.

Asset Class	$E[deg^*]$
U.S. Equities	1.48
Futures	1.33
World Indices	1.83
U.K. Equities	1.54
ETFs	1.39
Cryptocurrencies	2.42

Table 1: Mean optimal polynomial degree, deg^* , and 3-year Excess Return of the Volume-based prediction strategy less the buy-and-hold return over the same period, January 2022–December 2024.

if $\hat{R}_{CO,i,t+1} > 0.001$ **then** "Buy" asset i and hold the position until the end of the month $t+1$. The realized return for month $t+1$ is $R_{CO,i,t+1} = R_{i,t+1}$.
else
if $\hat{R}_{CO,i,t+1} < -0.001$ **then** "Short-sell" asset i and hold the position until the end of the month $t+1$. The realized return for month $t+1$ is $R_{CO,i,t+1} = -R_{i,t+1}$
else Do nothing. The realized return for month $t+1$ is $R_{CO,i,t+1} = 0$.
end if
end if

Next, we computed the "excess return" from the Conditional Optimization strategy for each asset under consideration. The excess return is the realized return under the CO strategy less the realized return on the asset for the same month:

$$R_{CO,i,t+1}^e = R_{CO,i,t+1} - R_{i,t+1} \quad (23)$$

The CO strategy outperforms the asset whenever it correctly predicts negative returns of the asset ahead of time and short-sells the asset instead of buying it. Table 2 shows the average excess returns across different asset classes.

Asset Class	$E[\text{Excess 3-Yr Return from Volume Strategy}]$
U.S. Equities	27.65%
Futures	20.87%
World Indices	14.34%
U.K. Equities	8.95%
ETFs	-133.52%
Cryptocurrencies	101.47%

Table 2: Mean optimal polynomial degree, deg^* , and 3-year Excess Return of the Volume-based prediction strategy less the buy-and-hold return over the same period, January 2022–December 2024.

As Table 2 shows, the CO strategy delivered positive returns for most asset classes. However, it produced negative average returns for ETFs.

The relationship between ETFs and volume has been discussed in Aldridge (2016), among others. One of the plausible explanations is the use of ETFs as easy liquidation assets: when a portfolio manager needs to liquidate assets in a hurry, perhaps due to a market crash, he can sell of very liquid ETFs first and then gradually work to rebalance his portfolio by selling his target assets and buying back the sold ETFs. Thus, ETFs often experience significantly negative volumes during sell-offs.

4.4 Portfolio Strategy Performance

Next, using $deg - 1$, we constructed the minimum-variance portfolios following the model in Section 3.2.2. Once again, we used the rolling window timeline shown in Figure 6. After we estimated the optimal polynomial degree deg_i for each instrument i , we predicted optimal allocations of each instrument in our sample minimum variance portfolio for N target assets:

$$w_{t+1}^* = \frac{\Sigma_t^{-1}e}{e\Sigma_t^{-1}e} \quad (24)$$

$$\Sigma_t = Var[\rho_t] \quad (25)$$

where ρ_t is an $N \times 1$ vector of asset returns (predicted and realized, more on this below), e is a vector of ones of length equal to the number of our target assets N , and $\Sigma_{CO,t}$ is the $N \times N$ variance-covariance matrix of ρ_t .

To calculate the optimal weights w_{t+1}^* , we considered the following cases:

- *Traditional Return Model*: realized returns on the assets $\{i\}$ at time t : $\rho_{it} = R_{it}$
- *Linear Prediction*: Changes in volume at time t for each asset i : $\rho_{it} = \Delta V_{it}$
- *Fully Non-Linear Prediction*: Changes in volume ΔV_{it} scaled by each asset's respective optimal polynomial degree deg_{it} : $\rho_{it} = \Delta V_{it}^{deg_{it}}$
- *Conditional Optimization Prediction*: Changes in volume ΔV_{it} scaled by each asset's *adjusted* optimal polynomial degree $deg_{it} - 1$: $\rho_{it} = \Delta V_{it}^{deg_{it}-1}$

Figure 7 shows resulting out-of-sample performance for each portfolio configuration. As the figures show, the Conditional Optimization technique outperformed the other portfolio models to deliver solid returns over time. Table 3 documents out-of-sample portfolio statistics for each methodology. The Conditional Optimization approach outperformed other models considered in terms of both annualized return and the Sharpe Ratio.

Figures in 8 show the respective allocations of the Min-var strategies out of the sample over time. As the figures show, traditional return-based minimum variance optimization has heavily favored the stablecoin USD ether (USDT-USD). At the same time, Conditional Optimization (CO) has been bullish on Bitcoin (BTC) since January 2023. However, since November 2024, the CO approach recommends dialing down on BTC and is giving a boost to the US equity markets (SPY) instead.

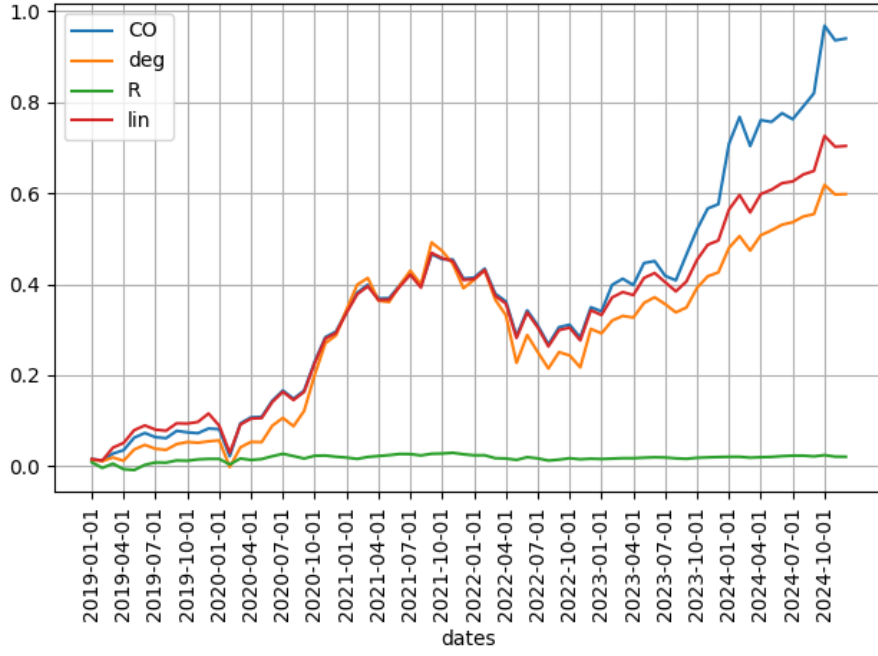


Figure 7: Minimum-Variance Portfolio Constructed with 1) Conditional Optimization (CO) prediction ($\rho_{it} = \Delta V_{it}^{deg_{it}-1}$) as well as other factors: 2) *deg*: $\rho_{it} = \Delta V_{it}^{deg_{it}}$, 3) linear *lin*: $\rho_{it} = \Delta V_{it}$ and 4) a traditional minimum-variance portfolio based on realized asset returns R_{it} .

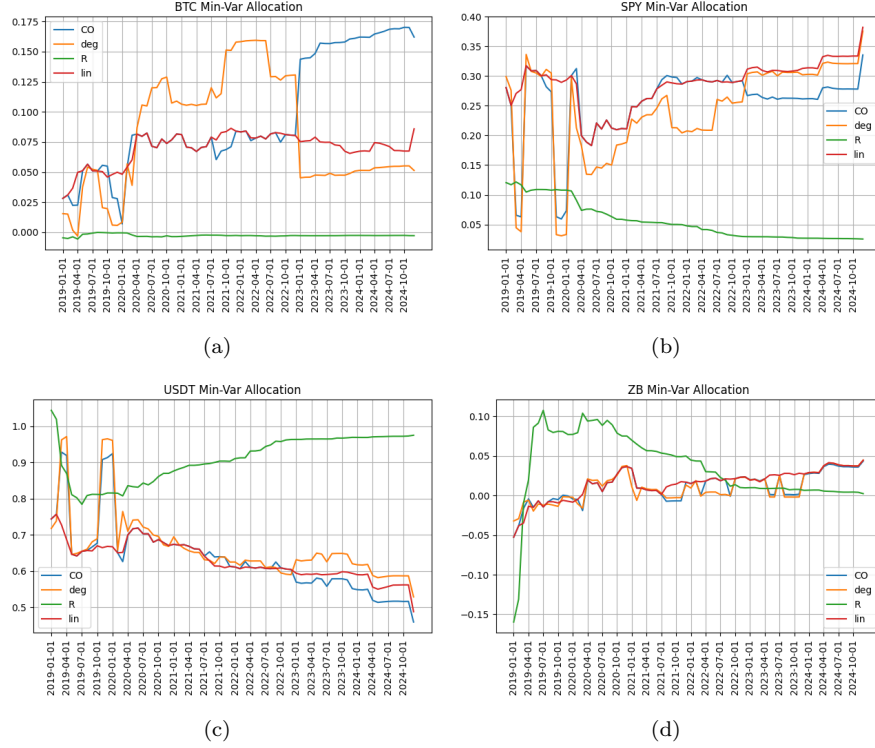


Figure 8: Min-Variance Portfolio Allocation for BTC-USD, SPY, ZB=F and USDT-USD computed using the following models: 1) Conditional Optimization (CO) prediction ($\rho_{it} = \Delta V_{it}^{deg_{it}-1}$), 2) *deg*: $\rho_{it} = \Delta V_{it}^{deg_{it}}$, 3) linear *lin*: $\rho_{it} = \Delta V_{it}$ and 4) a traditional minimum-variance portfolio based on realized asset returns R_{it} .

Portfolio Strategy	Annualized Portfolio Return	Annualized Sharpe Ratio
CO	11.56%	1.196
Nonlinear (full deg)	8.29%	0.870
Linear (lin)	9.27%	1.091
Traditional R-based	0.35%	0.232

Table 3: Portfolio performance comparison for min-variance portfolios constructed using 1) Conditional Optimization (CO), using $deg - 1$, 2) full-degree estimation using the highest polynomial degree deg , 3) linear approximation, and 4) traditional specification using returns R .

4.5 Robustness Checks

4.5.1 Determinants of Nonlinearity

Why do some assets have a linear and others a nonlinear relationship with returns? To find the cause of the degree of nonlinearity deg of returns R as a function of changes in volume ΔV , we run a series of regressions on various factors. Table 4 reports the results of the regression $deg_i = c_1 E[\Delta V_i] + c_2 \sigma[\Delta V_i] + \epsilon_i$ for different asset classes. The explanatory power of the model as measured by $Adj.R^2$ is particularly high for World Indices, ETFs and Cryptocurrencies.

As Table 4 shows, across all asset classes, the degree of nonlinearity is higher where the changes in volume are greater and more persistent. In other words, our model is picking up the volume momentum, a development consistent with Cong et al. (2020). The higher the momentum in volume, the higher the observed degree of non-linearity in $R \sim \Delta V^{deg}$.

Asset Class	$E[\Delta V]$	$\sigma[\Delta V]$	$Adj.R^2$
U.S. Equities	0.2427 (7.911)***	-0.0295 (-7.039)***	1.9%
Futures	0.2947 (1.596)	-0.0436 (-1.316)	7.2%
World Indices	129.6151 (1.858)*	-8.9078 (-1.674)*	28.1%
U.K. Equities	0.0074 (3.817)***	-0.0005 (-3.269)***	7.4%
ETFs	1.2681 (5.674)***	-0.1521 (-3.712)***	14.1%
Cryptocurrencies	2.7627 (0.420)	-0.1789 (-0.105)	13.0%

Table 4: Cross-sectional dependency of deg of nonlinearity in $R \sim \Delta V^{deg}$ on 1) Average change in volume, $E[\Delta V]$ and 2) Variation in the change in volume, $\sigma[\Delta V]$, for different asset classes. The table reports results of the regression $deg_i = c_1 E[\Delta V_i] + c_2 \sigma[\Delta V_i] + \epsilon_i$, where i is an individual asset considered within each asset class.

4.5.2 Trading Volume as a Cause of Price Momentum

Although we do not explicitly measure the momentum and its dependence on the trading volume, it is likely that the two are related, following Lee and

Swaminathan (2000). Furthermore, it is possible that the momentum is caused by the trading volume and can potentially explain the numerous momentum-based crypto strategies. Here, we investigate the dependency in detail.

We cannot rule out that the observed return momentum strategies are actually driven by the momentum in volume, as in Cong et al. (2020).

5 Directions for Future Research

Our research raises almost as many new questions as it answers. For example, are illiquid cryptocurrencies, stocks, ETFs and other financial instruments necessarily destined for the market graveyard? If so, how soon can we expect the deaths and does it make sense at all for large pension funds like Vanguard to ever invest in small-cap and microcap stocks? (We suspect that the answer to the latest question is a firm "no.")

Furthermore, much work remains to be done to investigate exactly how the prices of assets are related to the volume traded. What are the characteristics of the "volume-susceptible" assets versus the others? All of these questions will benefit academics, regulators, and practitioners in better understanding the nature of market forces.

6 Conclusions

Our analysis delivers a powerful new approach to crypto pricing, an area that has puzzled academics, regulators, and practitioners alike.

In addition, the presented results offer compelling new evidence for the valuation of all assets as transactional media. In line with Cong et al. (2020), the intrinsic quality of the assets is linked to their usage. The wider the transactional acceptance of the given asset as measured by the asset's traded volume, the higher the asset's market value.

Consistent with Cong et al. (2020), we find that newer assets such as cryptocurrencies have a polynomial relationship between asset prices and changes in volume traded. However, more established assets such as equities have a more linear relationship between the changes in volume and the future asset returns.

Estimating nonlinear models is a difficult process. To estimate the non-linear relationships between asset returns and volume changes, we deploy and test a recently proposed Conditional Optimization framework of Aldridge (2024) in real financial data. We find that its prediction for volume traded works for pricing assets. We tested and confirmed our predictions on monthly data. Our results show considerable out-of-sample performance.

Our conclusions about optimal portfolio holdings imply that new unproven assets are worth including in one's portfolio, but only as long as their user acceptance keeps growing over time. If the trading volume shrinks, the prices are likely to follow.

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